



Adventist University of Central Africa

SRS(Software requirements Systems)

Software Requirements Specification

Geospatial Disease Outbreak Mapper

****Version:**** 1.0

****Date:**** 19/11/2025

****Project:**** Geospatial Disease Outbreak Mapper

****Team:**** Group BC

****Status:**** Draft

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Introduction

1.1 Purpose

This document describes the requirements for the Geospatial Disease Outbreak Mapper system, which enables real-time tracking and mapping of disease outbreaks for rapid public health response.

1.2 Project Scope

The system provides:

- Mobile application for Community Health Workers (CHWs) to report disease cases
- Automatic GPS location capture
- Real-time outbreak mapping and visualization
- FHIR-compliant data standardization
- Central dashboard for health officials

1.3 Definitions, Acronyms, and Abbreviations

- ****CHW****: Community Health Worker
- ****FHIR****: Fast Healthcare Interoperability Resources
- ****GPS****: Global Positioning System
- ****SRS****: Software Requirements Specification
- ****UI****: User Interface

1.4 References

- Project Repository:
<https://gitlab.com/uwizeyemagnifique/final-project-group-bc>
- FHIR Specification: HL7 FHIR R4

1.5 Overview

This document is organized into sections describing overall system requirements, functional features, interface specifications, and design constraints.

2. Overall Description

2.1 Product Perspective

The system operates as a standalone mobile and web application that integrates with GPS services and follows FHIR healthcare data standards.

2.2 Product Functions

- Disease case reporting with auto-GPS capture
- Real-time outbreak mapping
- Data analytics and risk assessment
- Multi-user role management
- FHIR data export capabilities

2.3 User Characteristics

User Role	Responsibilities	Technical Skill
-----	-----	-----
Community Health Worker	Report disease cases, capture patient data	Basic mobile app usage
Public Health Official	Monitor outbreaks, view analytics, generate reports	Intermediate computer skills

2.4 Constraints

- Must comply with FHIR R4 standards
- GPS availability required for location services
- Mobile data connectivity for real-time updates

2.5 Assumptions and Dependencies

- CHWs have smartphones with GPS capability
- Health officials have internet access
- System assumes accurate GPS data

3. System Features

3.1 Disease Case Reporting

****Description:**** CHWs can report new disease cases through mobile application

****Priority:**** High

****Stimulus/Response:**** User submits form → System captures GPS and stores case

****Functional Requirements:****

- FR-001: System shall provide disease selection dropdown
- FR-002: System shall auto-capture GPS coordinates
- FR-003: System shall validate required fields
- FR-004: System shall convert data to FHIR Observation resource

3.2 Real-time Outbreak Mapping

****Description:**** Health officials can view live outbreak maps

****Priority:**** High

****Stimulus/Response:**** User requests map → System displays cases geographically

****Functional Requirements:****

- FR-005: System shall display cases on interactive map
- FR-006: System shall cluster nearby cases
- FR-007: System shall filter by disease type and date range

3.3 Risk Assessment

****Description:**** System automatically calculates outbreak risk levels

****Priority:**** Medium

****Stimulus/Response:**** New case reported → System updates risk assessment

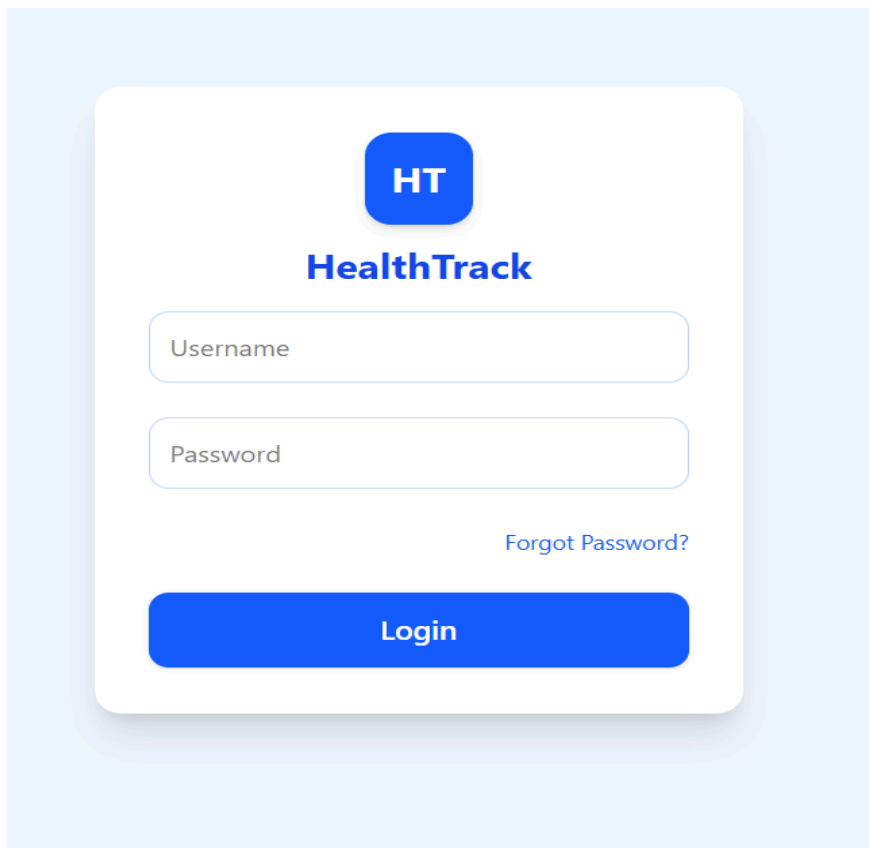
****Functional Requirements:****

- FR-008: System shall calculate risk based on case density
- FR-009: System shall generate RiskAssessment FHIR resources

4. External Interface Requirements

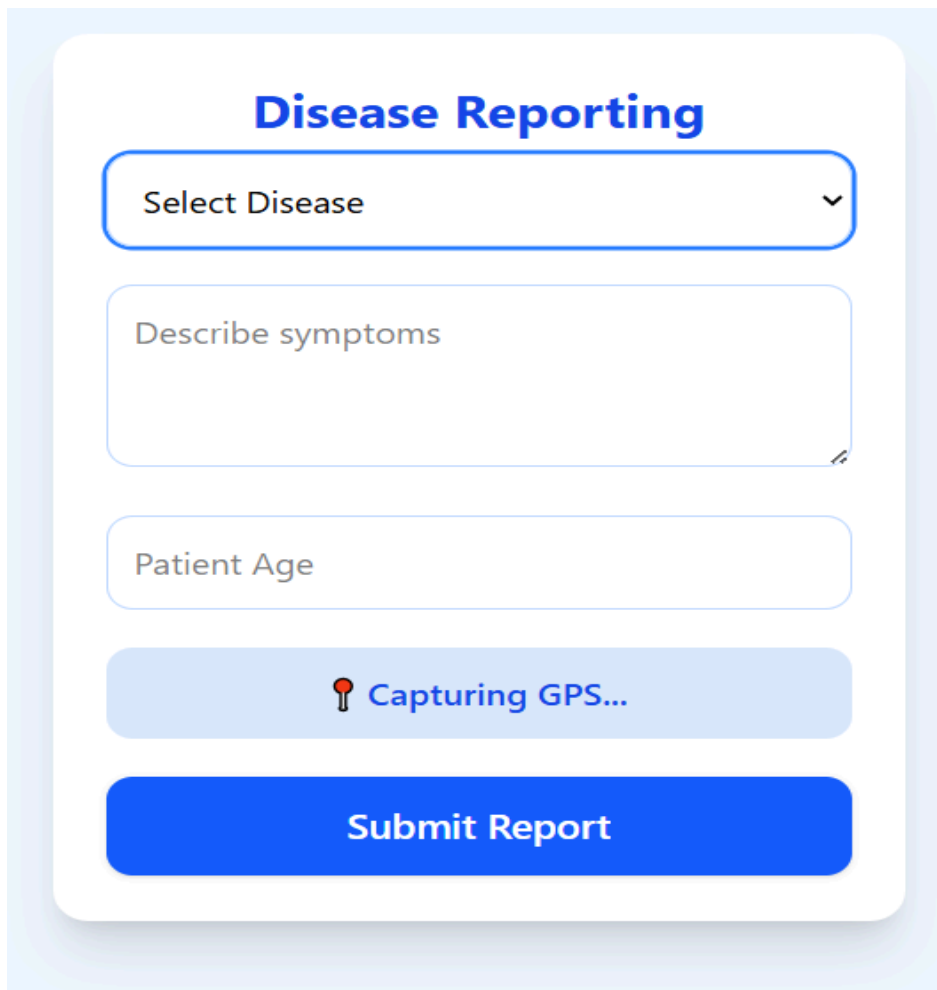
4.1 User Interfaces

4.1.1 CHW Mobile Application



![CHW Login Interface](ui-prototypes/chw-login.png)

Figure 1: Community Health Worker Login Screen

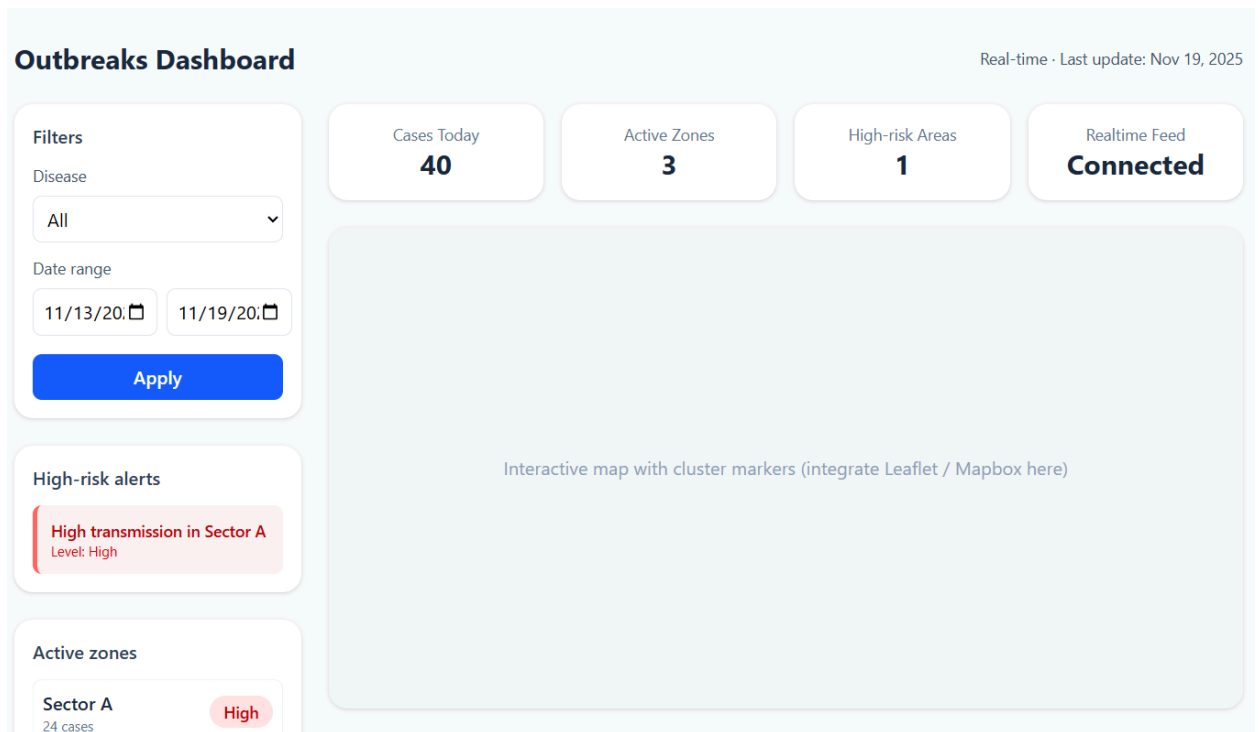


The image shows a UI prototype for a 'Disease Reporting' form. The form is contained within a white rounded rectangle with a blue border, set against a light blue background. At the top, the title 'Disease Reporting' is displayed in a bold, dark blue font. Below the title, there are four input fields: a dropdown menu labeled 'Select Disease' with a downward arrow, a text area labeled 'Describe symptoms' with a small pencil icon at the bottom right, a text input field labeled 'Patient Age', and a light blue button with a red location pin icon and the text 'Capturing GPS...'. At the bottom of the form is a large, solid blue button with the text 'Submit Report' in white.

![[Disease Reporting]](ui-prototypes/disease-reporting.png)

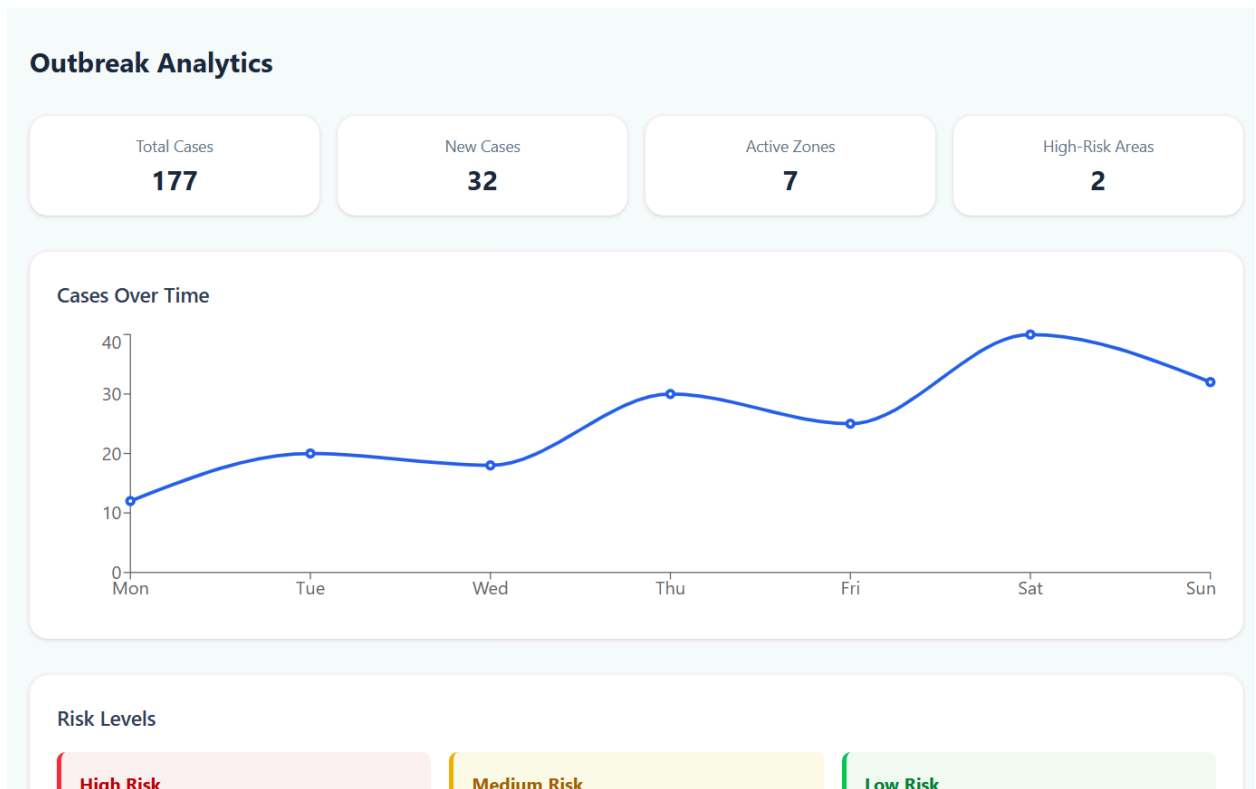
Figure 2: Disease Case Reporting Form

4.1.2 Health Official Dashboard



![Outbreak Dashboard](ui-prototypes/dashboard.png)

Figure 3: Real-time Outbreak Dashboard



![Analytics View](ui-prototypes/analytics.png)

Figure 4: Detailed Analytics View

4.2 Hardware Interfaces

- GPS receiver for location services
- Mobile device touch screen interface
- Network connectivity (WiFi/mobile data)

4.3 Software Interfaces

- Android/iOS mobile operating systems
- FHIR R4 compliant backend services
- GPS location services API

4.4 Communications Interfaces

- HTTPS/REST API for data transmission
- JSON data format for FHIR resources

- Real-time WebSocket connections for live updates

5. Non-Functional Requirements

5.1 Performance Requirements

- Map should load within 3 seconds
- Case reports should submit within 5 seconds
- System should support 1000+ concurrent users

5.2 Safety Requirements

- No single point of failure for emergency reporting
- Data backup and recovery procedures

5.3 Security Requirements

- All data encrypted in transit (HTTPS/TLS)
- Role-based access control
- User authentication required for all operations
- Audit logs for all data modifications

5.4 Software Quality Attributes

- ****Availability:**** 99.5% uptime requirement
- ****Reliability:**** Offline capability for mobile app
- ****Maintainability:**** Modular architecture with clear documentation
- ****Portability:**** Support for Android and iOS platforms

6. Other Requirements

6.1 Database Requirements

- FHIR-compliant data storage
- Spatial indexing for geographic queries
- Audit trail for all data changes

6.2 Operations Requirements

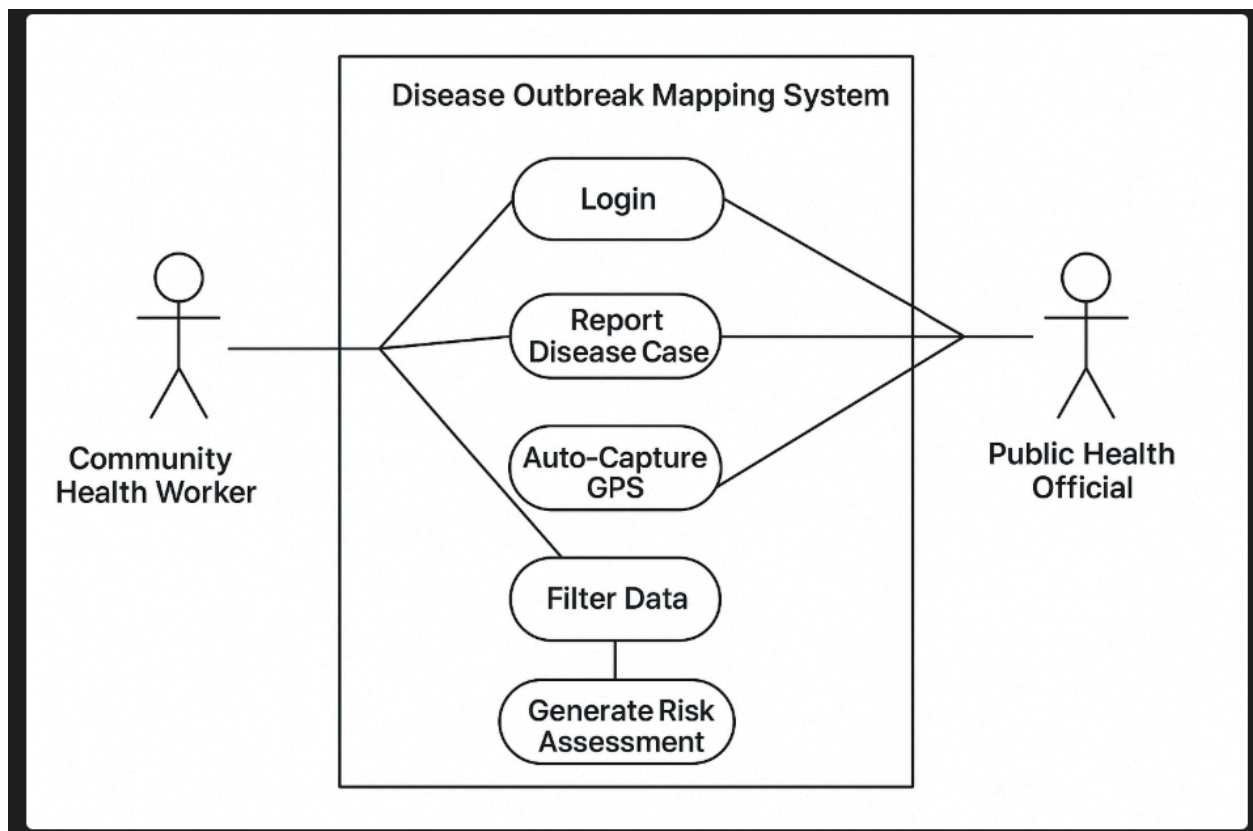
- Automated backup procedures
- Monitoring and alerting system
- Regular security updates

6.3 Appended Requirements

- Compliance with local health data regulations
- Accessibility features for users with disabilities

7. Appendices

7.1 Use Case Diagrams



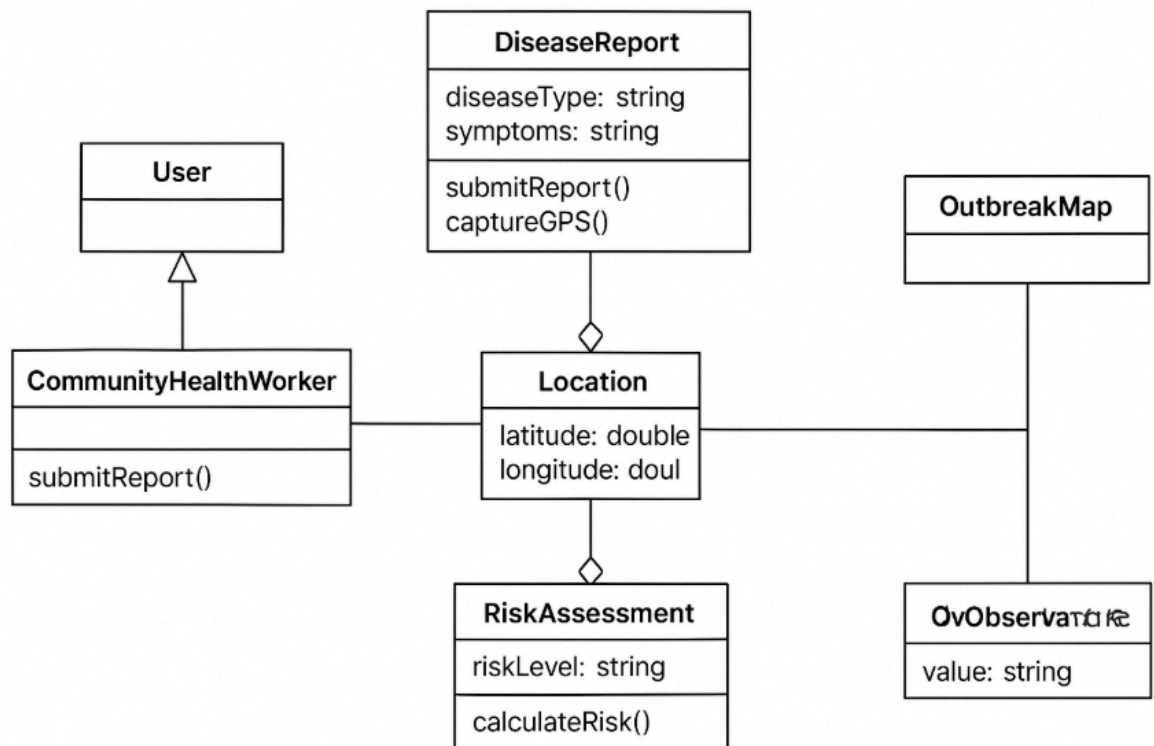
![Use Case Diagram](diagrams/use-case-diagram.png)

Figure 5: System Use Case Diagram

****Actor Descriptions:****

- ****Community Health Worker:**** Reports disease cases from field locations
- ****Public Health Official:**** Monitors outbreaks and views analytics
- ****System:**** Automated processes for data handling and notifications

7.2 Class Diagrams

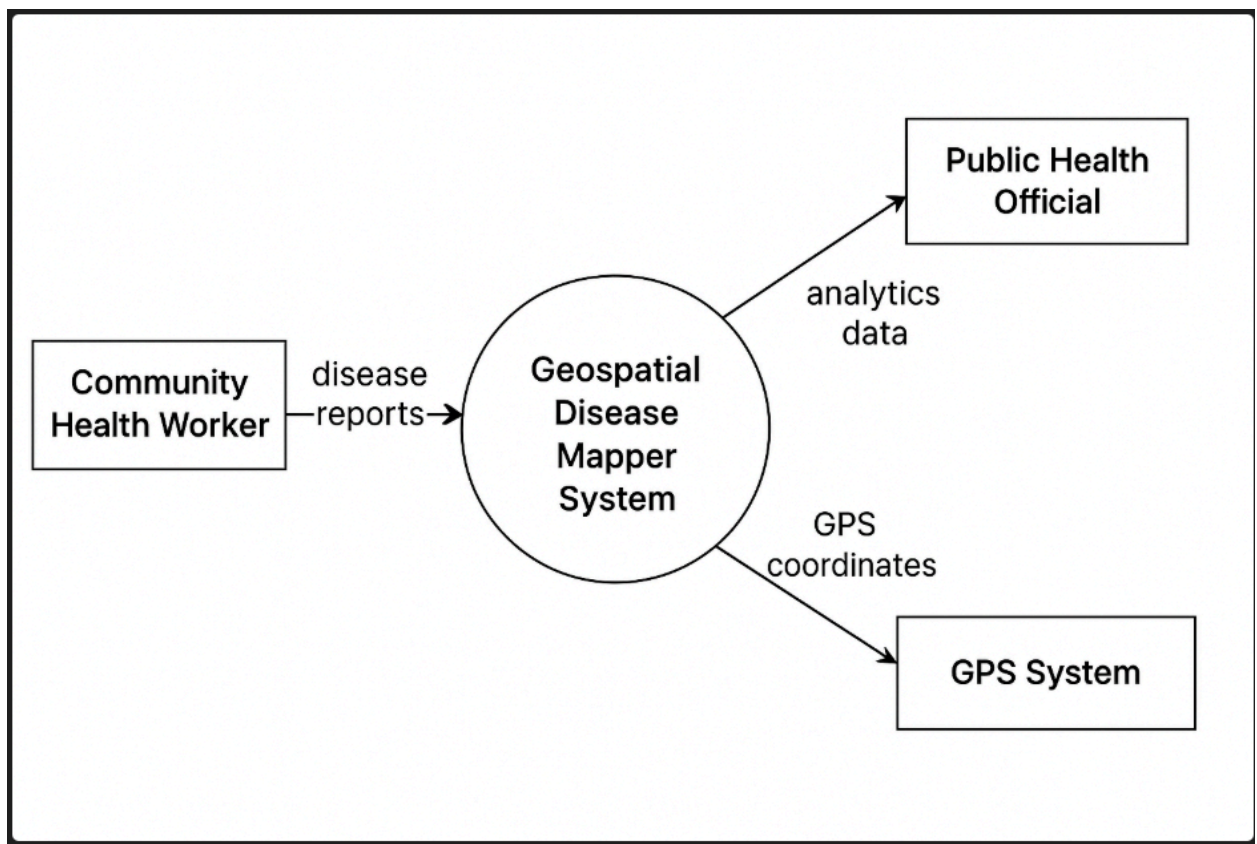


![Class Diagram](diagrams/class-diagram.png)

Figure 6: System Class Diagram

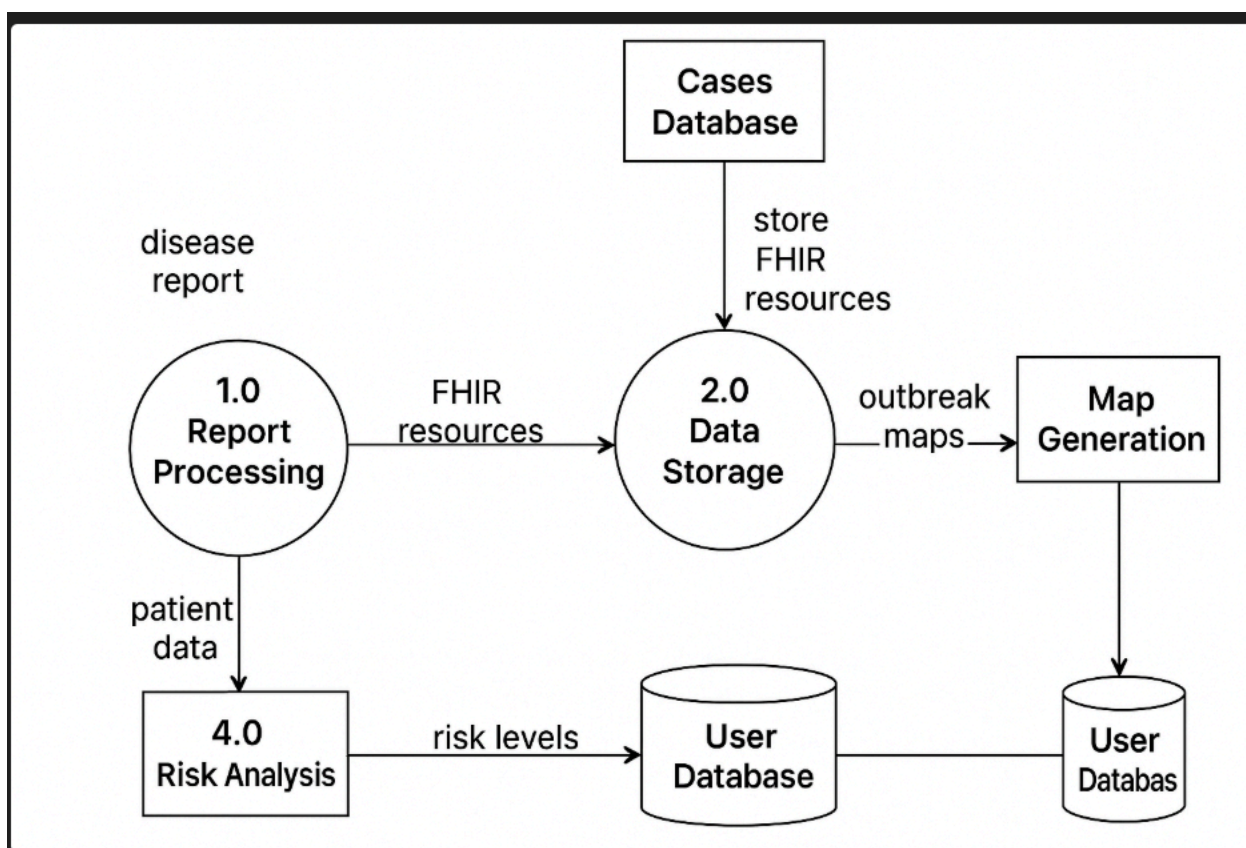
****Key Classes:****

- `DiseaseReport`: Core entity for disease case reporting
- `Location`: GPS coordinates and spatial data
- `User`: Base class for system users
- `Observation`: FHIR resource for clinical data

7.3 Data Flow Diagrams

![[Data Flow Diagram Level 0]](diagrams/dfd-level0.png)

Figure 7: Context Diagram



![[Data Flow Diagram Level 1](diagrams/dfd-level1.png)]

Figure 8: Level 1 DFD

7.4 FHIR Resource Mapping

System Component	FHIR Resource	Purpose
-----	-----	-----
Disease Case	Observation	Clinical findings and diagnosis
Patient Encounter	Encounter	Interaction between CHW and patient
Location Data	Location	GPS coordinates and place information
Risk Analysis	RiskAssessment	Outbreak risk calculations
Analytics	MeasureReport	Aggregated data summaries

7.5 Revision History

Version	Date	Author	Changes
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1.0	19/11/2025	Group BC	Initial SRS version

Approval

Prepared by:

Group BC Team